

Notes: Best, Cloyne, Ilzetzki, and Kleven (RES, 2020)

Estimating the Elasticity of Intertemporal Substitution Using Mortgage Notches

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1 Big picture

- **Question.** How responsive are households to changes in the intertemporal price of consumption? In standard dynamic models, this response is summarized by the elasticity of intertemporal substitution (EIS), denoted σ .
- **Empirical innovation.** The paper exploits a quasi-experimental source of interest-rate variation in the U.K. mortgage market: discrete mortgage-rate jumps at loan-to-value (LTV) thresholds. These jumps create sharp incentives to reduce borrowing just enough to fall below an LTV threshold.
- **Setting.** U.K. mortgage products are priced in LTV bands. Crossing thresholds such as 60%, 70%, 75%, 80%, and 85% raises the interest rate on the entire loan, not only on marginal borrowing.
- **Main identifying pattern.** The observed LTV distribution has sharp bunching just below notches and missing mass just above notches. This reveals that households respond to interest-rate incentives.
- **Why refinancers matter.** The analysis focuses on refinancers, because housing choices are predetermined. This isolates borrowing and intertemporal consumption choices from housing demand.
- **Method.** The paper translates bunching moments into a structural EIS using two models: a transparent two-period model and a richer stochastic lifecycle model.
- **Main result.** The average EIS is small, roughly 0.1. The result is stable across notches, structural specifications, friction corrections, risk preferences, liquidity choice, present bias, and other robustness checks.
- **Reduced-form versus structural.** The reduced-form elasticity of borrowing with respect to interest rates is around 0.5, but this does not imply a high EIS. Most of the borrowing response comes from the wealth effect of higher interest costs, not strong intertemporal substitution.
- **Why the result is surprising.** Large observed bunching may initially look like evidence of high responsiveness. The paper shows the opposite: given how large the notched incentives are, high EIS values would generate much more bunching and multiple-notch jumps.
- **Economic takeaway.** Households are willing to reduce leverage to avoid mortgage-rate notches, but the implied willingness to substitute consumption intertemporally is very low. This challenges calibrations that use EIS values near or above one.

2 Incremental contribution of the paper

2.1 Contribution relative to the EIS literature

- The traditional EIS literature estimates Euler equations using aggregate time-series variation or survey-based micro consumption data.
- Those estimates are sensitive to measurement error, weak instruments, asset-market participation, aggregation, preference assumptions, and endogenous interest-rate movements.
- This paper instead uses cross-sectional, point-in-time, quasi-experimental variation in borrowing rates generated by mortgage-rate notches.
- The interest-rate discontinuity is localized and institutional: borrowers face different mortgage rates when their LTV crosses a threshold.
- The result is a clean empirical moment—bunching below thresholds—that can be connected to structural intertemporal preferences.

Interpretation: The paper’s incremental contribution is not simply another EIS estimate. It supplies a new identification strategy for the EIS that avoids the standard reliance on gradual and endogenous time-series interest-rate movements.

2.2 Contribution relative to mortgage-demand and borrowing-elasticity studies

- A separate literature estimates reduced-form borrowing elasticities with respect to after-tax or regulated interest rates.
- Such elasticities are useful but do not directly reveal the EIS because borrowing responses mix substitution effects, wealth effects, collateral constraints, and household balance-sheet conditions.
- The paper explicitly derives the mapping between the structural EIS and a reduced-form borrowing elasticity.
- This mapping shows that the same reduced-form borrowing elasticity can correspond to many different EIS values, depending on loan-to-wealth ratios and other state variables.
- The empirical estimate of the borrowing elasticity is about 0.5, while the structural EIS is about 0.1.

Interpretation: The paper clarifies why a mortgage-demand elasticity is not the same object as the EIS. A large borrowing response can be generated largely by the wealth effect of a higher mortgage payment rather than by strong willingness to shift consumption over time.

2.3 Contribution relative to bunching methods

- Most bunching applications estimate static reduced-form elasticities at tax kinks or notches.
- This paper applies bunching to an inherently dynamic choice: mortgage borrowing today versus future consumption.
- The paper builds a structural mapping from bunching to the EIS rather than stopping at a reduced-form elasticity.
- It also extends the Kleven–Waseem idea of using dominated regions above notches to measure optimization frictions.
- In this mortgage setting, the dominated region must be derived structurally because locating just above an LTV notch gives slightly more current consumption but much worse future consumption.

Interpretation: The incremental methodological contribution is to show how quasi-experimental bunching moments can identify deep preference parameters in a dynamic model, while also using the missing mass above notches to discipline optimization frictions.

2.4 Contribution relative to structural lifecycle models

- Structural lifecycle models can fit many behaviors, but their conclusions often depend on assumptions about risk, liquidity, beliefs, assets, and household heterogeneity.
- The authors begin with a simple two-period model, then build a richer stochastic lifecycle model with housing, liquid assets, uncertainty, Epstein–Zin preferences, present bias, risk aversion, and refinancing costs.
- The two models deliver nearly the same answer.
- This is important because it shows that the small EIS is not an artefact of the simple model.
- The empirical shape of the LTV distribution, especially the absence of multiple-notch jumps, strongly disciplines the structural parameter.

Interpretation: The structural contribution is robustness-through-comparison: the transparent simple model and the richer lifecycle model agree, so the result is not driven by black-box assumptions.

2.5 Substantive policy contribution

- The EIS is central in macroeconomics, public finance, household finance, and asset pricing.
- Many calibration exercises rely on larger EIS values, often near one.
- The paper provides quasi-experimental evidence that the household EIS is much smaller.
- Low EIS implies limited substitution of consumption across time in response to interest rates.
- For monetary and fiscal policy, this means that intertemporal substitution channels may be weaker than models with high EIS suggest.

Interpretation: The policy relevance is that interest-rate changes may affect household behavior more through wealth/cash-flow effects than through classic substitution effects.

3 Data and empirical setting

3.1 U.K. mortgage-market features

- The U.K. mortgage market has discrete interest-rate notches at LTV thresholds.
- The relevant thresholds are 60%, 70%, 75%, 80%, and 85%. The 90% threshold is visible in the raw distribution but is not central to the estimation because post-crisis mortgage supply above 90% was limited.
- When a borrower crosses a threshold, the interest rate rises on the entire mortgage balance.
- This makes the notch a large discrete incentive rather than a small marginal kink.
- Mortgage products typically offer a low initial rate for 2–5 years, followed by a much higher reset rate.
- Borrowers have strong incentives to refinance near the reset date, while prepayment charges discourage refinancing too early.

Interpretation: The market structure creates a clean decision margin: at refinancing, borrowers choose how much to borrow, and the interest-rate schedule makes certain LTV thresholds discretely costly.

3.2 Data sources

- The main data are the Financial Conduct Authority Product Sales Database (PSD), covering the universe of regulated U.K. mortgage originations.
- The paper uses mortgage data from 2008 to 2014.
- The data include loan size, property value, LTV, interest rate, mortgage term, reset-date information, fixed versus variable rate status, repayment type, borrower income, age, joint-income status, and refinance indicators.
- The authors supplement PSD data with MoneyFacts product data, which contain product-level mortgage fees and interest-rate schedules.
- The MoneyFacts match is important because origination fees can be rolled into the loan and mechanically distort the observed LTV relative to the LTV used by banks for pricing.

Interpretation: These data allow the authors to observe both the household’s actual LTV choice and the bank/product/time-specific interest-rate schedule that generated the incentive.

3.3 Why focus on refinancers

- In purchase mortgages, LTV can reflect both mortgage borrowing and housing demand.
- In refinancing, the house is already chosen.
- Therefore, changes in LTV primarily reflect borrowing, equity extraction, and repayment choices.
- This is essential for estimating intertemporal substitution: the authors want to know how households trade current resources against future mortgage payments, not how they choose house size.
- The refinancer panel also allows construction of a passive LTV based on the previous mortgage, amortization, and new property valuation.

Interpretation: Refinancers make the empirical design much cleaner because they allow the paper to isolate the leverage/borrowing margin from the housing-demand margin.

3.4 Descriptive sample

- The full sample contains about 3.05 million mortgage observations.
- The refinancer sample contains about 1.96 million observations.
- The refinancer panel used for counterfactual construction contains about 647,000 observations.
- Summary statistics are stable across samples: interest rates, loan sizes, property values, borrower age, and income are broadly similar.
- The refinancer panel has an average interest rate of 4.06%, loan size about £145,401, property value about £262,304, LTV around 58.8%, and gross income around £57,483.

Interpretation: The estimation sample is not a tiny special group. It remains broadly similar to the larger mortgage population while providing the panel information needed for counterfactuals.

4 Interest-rate notches and counterfactual LTV distributions

4.1 Estimating interest-rate jumps

The authors estimate conditional interest-rate schedules using a non-parametric step function in LTV:

$$r_i = f(LTV_i) + \beta_1 bank_i + \beta_2 variability_i \otimes duration_i \otimes month_i + \beta_3 repayment_i + \beta_4 term_i + \nu_i. \quad (1)$$

- r_i is the nominal mortgage interest rate.
- $f(LTV_i)$ is a step function in 0.25 percentage-point LTV bins.
- Bank fixed effects absorb lender-specific pricing differences.
- Product/time controls absorb fixed versus variable products, initial-rate duration, origination month, repayment type, and mortgage term.
- The coefficient changes in $f(\cdot)$ at thresholds identify the conditional interest-rate jumps.

Interpretation: The regression compares borrowers within bank/product/time cells, so the estimated notch is not driven by borrowers sorting into different banks or products. It measures the price change faced by a borrower on a given product when crossing the LTV threshold.

4.2 Estimated notch sizes

- The interest schedule is approximately flat between thresholds.
- It jumps sharply at 60%, 70%, 75%, 80%, and 85% LTV.
- The 80% and 85% thresholds are especially large, with jumps near half a percentage point in the global interest-rate schedule.
- The paper later uses local estimates for refinancers when computing bunching-based elasticities.

Interpretation: Flatness between thresholds strengthens the design. If rates changed smoothly with LTV, bunching might be due to continuous pricing or risk gradients. The discrete jumps make the quasi-experiment more credible.

4.3 Why a special counterfactual is needed

- Standard bunching designs often fit a polynomial around one notch while excluding the bunching area.
- This does not work well here because the LTV distribution is affected by multiple nearby notches.
- Observations near one threshold may be distorted by another threshold.
- The authors therefore use the panel structure of refinancing to construct a passive LTV distribution.

Interpretation: The paper’s counterfactual contribution is important: it avoids treating distorted parts of the observed LTV distribution as if they were undistorted controls.

4.4 Passive LTV and adjusted counterfactual

- Passive LTV is the LTV a refinancer would have if she rolled over the prior mortgage balance, accounting for amortization and property value.
- The passive distribution is smooth and does not bunch at notches.
- But actual refinancers may systematically extract or inject equity, so passive LTV is adjusted using the observed equity-extraction behavior of non-bunchers.
- The resulting counterfactual distribution is the estimated distribution that would arise without the mortgage-rate notches.

Interpretation: The counterfactual captures what the distribution would look like absent notch incentives, not merely what it looked like under the previous mortgage.

5 Simple structural model

5.1 Preferences, budget constraints, and Euler equation

Households live for two periods and value non-housing consumption with constant EIS preferences. Housing is fixed because the sample is refinancers. Let λ denote LTV, P_0H housing value, W_0 initial net wealth, y_t income, R the gross mortgage rate, and d depreciation. The period budgets are:

$$c_0 = y_0 + W_0 - (1 - \lambda)P_0H, \quad (2)$$

$$c_1 = y_1 - R\lambda P_0H + (1 - d)P_1H. \quad (3)$$

The Euler equation is:

$$c_1 = (\delta R)^\sigma c_0. \quad (4)$$

Interpretation: Increasing λ raises current resources because the household borrows more, but lowers future consumption because mortgage repayment is higher. The EIS governs the curvature of the trade-off between current and future consumption.

5.2 How a notch creates bunching

- Without a notch, a household with a counterfactual LTV just above λ^* would choose an interior LTV above the threshold.
- With a notch, the interest rate jumps from R to $R + \Delta R$ above λ^* .
- Some households that would otherwise locate above λ^* reduce LTV to exactly λ^* .
- The marginal buncher is indifferent between locating at the notch and choosing the best interior LTV above the notch.
- The distance $\Delta\lambda$ from the notch to the marginal buncher's counterfactual LTV is inferred from the excess bunching mass.

Interpretation: Larger EIS implies greater willingness to give up current consumption in order to avoid a higher future mortgage cost. Thus, for a given interest-rate jump, higher EIS predicts a larger LTV response and more bunching.

5.3 Estimating indifference equation

The paper derives indirect utilities for locating at the notch and for locating at the best interior point. The EIS solves:

$$F(\sigma, \delta, \Delta\lambda, \Delta R, x) = V^N(\sigma, \delta, \Delta\lambda, x) - V^I(\sigma, \delta, \Delta\lambda, \Delta R, x) = 0. \quad (5)$$

The vector x contains calibrated and observed objects such as R , λ^* , future income relative to housing wealth, and house price growth net of depreciation.

Interpretation: The bunching moment $\Delta\lambda$ and interest-rate jump ΔR are empirical. The equation asks what curvature σ makes the marginal buncher exactly indifferent between bunching and not bunching.

5.4 Reduced-form borrowing elasticity

The paper also derives the relationship between the EIS and the reduced-form elasticity of borrowing with respect to the interest rate:

$$\varepsilon = -\frac{\partial \log \lambda}{\partial \log R} = \frac{\sigma(\delta R)^\sigma + R}{(\delta R)^\sigma + R} - \frac{\sigma(\delta R)^\sigma \times LTV}{1 + (\delta R)^\sigma \times LTV}. \quad (6)$$

- LTV is a loan-to-wealth type ratio.
- The elasticity depends on the EIS, discount factor, interest rate, and the household's loan-to-wealth ratio.
- When σ is very small, a positive borrowing response remains because higher rates reduce lifetime wealth.

Interpretation: This is central for interpretation. A reduced-form borrowing elasticity around 0.5 does not necessarily indicate a high EIS; it can largely reflect the wealth effect of a higher mortgage cost.

5.5 Optimization frictions and dominated regions

- Some households may remain just above a notch due to inattention, misunderstanding, switching costs, or other frictions.
- If those non-optimizers are counted as optimizers, the EIS estimate would be biased downward.
- The paper adapts the Kleven–Waseem dominated-region logic to the mortgage setting.
- The key idea is that for any non-negative EIS, a sufficiently close point above the notch is dominated by moving down to the notch.
- The mass in this dominated region estimates the fraction of non-optimizers.

The dominated range is characterized by:

$$\lambda^* + \Delta\lambda \in \left(\lambda^*, \left(1 + \frac{\Delta R}{R+1} \right) \lambda^* \right). \quad (7)$$

Interpretation: The missing mass and dominated-region mass let the paper correct for optimization frictions without relying only on parametric assumptions about inattention.

5.6 Simulation logic

- The authors simulate LTV distributions under different EIS values.
- If $\sigma = 1$, the model predicts very large responses: many households jump across multiple notches, leaving holes between thresholds.
- The observed data do not show such behavior.
- A very small EIS, around 0.06–0.12, fits the shape of the LTV distribution much better.

Interpretation: The simulations make the identification visually intuitive: high-EIS households would not merely bunch a little; they would radically reorganize the LTV distribution.

6 Empirical estimates from the simple model

6.1 Pooled notch estimation

- The authors normalize LTV around each threshold and stack the five notches into a pooled notch.
- The actual distribution has a large spike exactly below the notch and a hole above the notch.
- The pooled interest-rate jump is about 0.25 percentage points.
- Raw normalized bunching is about five times the counterfactual density at the notch.
- After adjusting for round-number bunching, the normalized bunching measure is about 4.45.
- The hole extends about 3.5 percentage points above the notch.

Interpretation: The pooled notch provides a visually clear estimate of how far the marginal buncher moved and how much of the nearby mass appears non-optimizing.

6.2 Individual notch estimation

- The individual notch graphs show the 70%, 75%, 80%, and 85% thresholds separately.
- Bunching rises with the size of the interest-rate jump.
- The 70% notch has a smaller interest-rate jump and less bunching.
- The 80% and 85% notches have larger jumps and larger bunching.
- This monotonicity is an important validation of the causal interpretation.

Interpretation: Borrowers respond more when the interest-rate incentive is larger, which supports the interpretation that bunching is driven by mortgage pricing rather than by unrelated round-number preferences.

6.3 Table 2: From bunching to EIS

- The average interest rate below the pooled notch is about 3.42%.
- The pooled interest-rate jump is about 0.25 percentage points.
- The pooled bunching measure is 4.45 before friction adjustment and 6.30 after friction adjustment.
- The implied average LTV response is about 1.93 percentage points after adjustment.
- The pooled structural EIS estimate is 0.07 in the simple model.
- Across individual notches, the EIS ranges from 0.03 to 0.17.
- The reduced-form borrowing elasticity is stable around 0.55.

Interpretation: The key message is that observed bunching is large in raw terms but small relative to the incentive created by the notch. Hence the structural EIS is low.

6.4 Table 3: Bounds from optimization frictions

- Without friction adjustment, the pooled EIS is about 0.05.
- Using the baseline dominated-region adjustment, the pooled EIS is about 0.09.
- Under the extreme assumption that all mass in the hole is due to friction, the pooled EIS rises to about 0.30.
- The upper-bound scenario is intentionally conservative.

Interpretation: Even aggressive friction assumptions do not produce an EIS near one. Optimization frictions matter, but they cannot reconcile the data with large intertemporal substitution.

6.5 Table 4: Heterogeneity

- EIS estimates are modestly larger for older households.
- They are somewhat larger for households with higher house price growth and larger passive interest-rate changes.
- They are lower for higher-income households in the quartile comparison.
- Across all groups, estimates remain small, generally between about 0.02 and 0.15.

Interpretation: The low EIS is not hiding strong responses among a large subgroup. Heterogeneity exists, but it is quantitatively limited.

7 Full stochastic lifecycle model

7.1 Why the full model is needed

- The simple model assumes two periods, perfect foresight, no liquid assets, and no uncertainty.
- A richer model is needed to check whether the EIS result survives realistic household features.
- The full model allows for housing, liquid assets, stochastic income, stochastic house prices, mortgage refinancing, amortization, reset rates, origination fees, bequest motives, and Epstein–Zin preferences.
- This separates risk aversion from the EIS.

Interpretation: The lifecycle model is a robustness engine. It asks whether realistic household balance-sheet channels can explain the bunching with a high EIS.

7.2 Preferences in the full model

The recursive utility is:

$$V_t = \left((c_t^\alpha H_{t+1}^{1-\alpha})^{\frac{\sigma-1}{\sigma}} + \delta \left(E_t[V_{t+1}^{1-\gamma}] \right)^{\frac{1}{1-\gamma} \frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}. \quad (8)$$

The terminal bequest/retirement value is:

$$V_{T+1} = \Gamma W_{T+1}. \quad (9)$$

Interpretation: Epstein–Zin preferences matter because they allow the authors to vary risk aversion without mechanically changing the EIS.

7.3 Budget constraint and mortgage structure

The household budget constraint in the full model is:

$$c_t = y_t + (1 - \pi_t)L_t - L_{t+1} + P_t((1 - d)H_t - H_{t+1}) \quad (10)$$

$$+ D_{t+1} - R_t D_t - \Omega I_t^R. \quad (11)$$

- L_t is liquid asset holdings.
- D_t is mortgage debt.
- H_t is housing.
- R_t is the real gross mortgage rate.
- Ω is a fixed mortgage origination fee.
- I_t^R indicates refinancing.
- The mortgage spread is a step function of the LTV at origination.

Interpretation: Liquid assets provide another way to bunch. A household can reduce LTV not only by cutting current consumption but also by drawing down liquid assets. This generally makes a given amount of bunching consistent with an even lower EIS.

7.4 Full-model estimation

- The full model is solved numerically.
- Parameters other than the EIS are calibrated to data or values from the literature.
- The EIS is estimated by matching friction-adjusted bunching moments.
- The empirical targets include bunching, friction adjustment, and LTV responses at individual notches.
- The model allows repeated refinancing episodes, so households anticipate future notch incentives.

Interpretation: The full model is less transparent but more realistic. It checks whether dynamic portfolio and liquidity choices alter the simple-model conclusion.

7.5 Full-model results

- The average EIS from the full structural model is 0.08.
- Individual estimates are 0.05 at 60%, 0.04 at 70%, 0.11 at 75%, 0.11 at 80%, and 0.28 at 85%.
- The 85% estimate is larger because households near that notch may face stronger liquidity considerations.
- Despite this, the overall conclusion remains that EIS is small.
- The full-model estimate is very close to the simple-model estimate.

Interpretation: The agreement between the simple and full models is one of the paper’s strongest results. It suggests that the empirical bunching moments themselves tightly discipline the EIS.

7.6 Robustness checks

- Changing the discount factor leaves the EIS estimate moderate.
- Allowing present bias also does not generate a high EIS.
- Changing risk aversion over a wide range has limited impact.
- Shifting expected future interest rates, house price trends, house price variance, income profiles, unemployment risk, and replacement rates does not overturn the result.
- The baseline estimate in Table 6 is generally around 0.08.

Interpretation: The robustness table reinforces the central message: the low EIS is not a fragile calibration result.

8 Identification and credibility

8.1 What identifies the causal interest-rate response

- Interest-rate notches generate discrete incentives at known LTV thresholds.
- The thresholds are institutional and highly salient.
- Within bank/product/time cells, rates are offered as a menu rather than individually negotiated.
- The rate schedule is approximately flat between notches and jumps at thresholds.
- Bunching occurs exactly where the incentives predict.

Interpretation: The main causal object is the local LTV response to discrete changes in mortgage rates. The sharpness of bunching and the flatness between notches support the design.

8.2 Why selection into refinancing is less problematic

- Refinancing tends to occur when the initial mortgage-rate period ends.
- Prepayment penalties discourage earlier refinancing.
- Reset rates encourage refinancing once the initial rate expires.
- This institutional timing reduces concern that households choose refinance timing strategically around LTV notches.

Interpretation: The design is stronger because the timing of the refinancing decision is partly pinned down by contract structure, while the LTV choice remains the key margin.

8.3 Key assumptions

- Default risk and other loan-quality dimensions are smooth around thresholds absent the notches.
- The counterfactual LTV distribution can be constructed from passive LTV plus equity-extraction adjustment.
- Round-number bunching can be netted out using no-notch observations.
- The dominated-region approach provides a reasonable measure of optimization frictions.
- Refinancers' LTV responses map into intertemporal consumption/liquidity choices through the model.

Interpretation: The assumptions are transparent and directly tied to the institutional setting. The strongest concern is not whether households respond to notches, but how to translate that response into a structural EIS.

8.4 Main limitations

- The design identifies behavior of U.K. mortgage refinancers, not all households in all settings.
- Mortgage borrowing is a specific intertemporal margin and may not equal the EIS relevant for all consumption choices.
- The structural translation requires assumptions about expectations, wealth, housing, and liquidity.
- The dominated-region friction adjustment is informative but cannot perfectly classify all non-optimizers.
- The data do not directly observe non-housing assets and liabilities, so the full model must calibrate them.

Interpretation: The paper is best read as strong evidence that a clean household borrowing margin implies a low EIS, rather than as a universal estimate for every intertemporal choice margin.

9 Tables and figures

9.1 Figures 1–2: raw bunching and high-EIS counterfactual

Read:

- Figure 1 shows large spikes below 60%, 70%, 75%, 80%, 85%, and 90% LTV thresholds.
- The spikes are accompanied by missing mass above thresholds.
- Figure 2 compares the actual distribution to a simulated distribution under EIS $\sigma = 1$.
- The simulated high-EIS distribution has far too much bunching at low notches and too little at high notches.
- Under $\sigma = 1$, borrowers would often jump across multiple notches, which is not observed.

Economic message: The raw distribution proves that borrowers respond to mortgage-rate incentives. The simulation shows that the observed response is too small to support standard high EIS values.

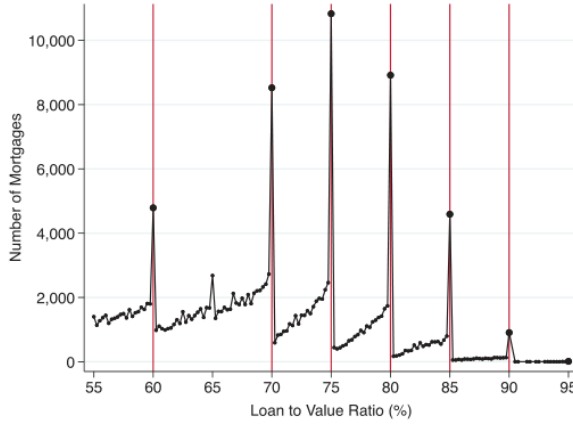


FIGURE 1

Observed LTV distribution among U.K. refinancers

Shows the observed distribution of LTV ratios among refinancers in the U.K. between 2008–14. Thresholds of 60%, 70%, 75%, 80%, 85%, and 90% (depicted by vertical lines).

(a) Observed LTV distribution among U.K. refinancers.

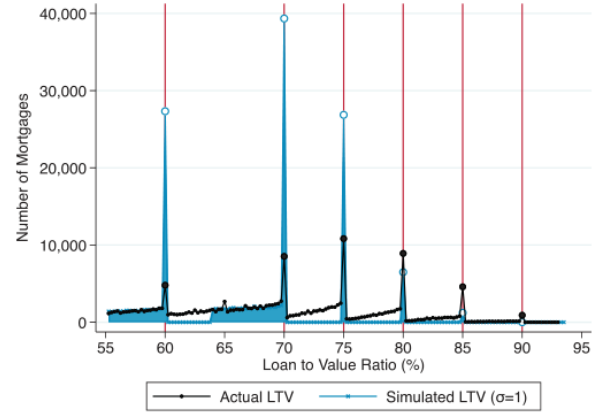


FIGURE 2

Observed versus simulated LTV distributions under an EIS = 1

Compares the observed LTV distribution (black markers) to a simulated LTV distribution (lighter blue line).

(b) Observed versus simulated distribution under EIS = 1.

9.2 Table 1 and equation (2.1): sample and interest-rate schedule estimation

Read:

- Table 1 reports summary statistics for the full sample, refinancer sample, and refinancer panel.
- The refinancer panel contains 647,192 observations.
- The average LTV in the refinancer panel is 58.8%.
- Equation (2.1) shows how interest-rate jumps are estimated using an LTV step function plus bank, product, time, repayment, and term controls.

Economic message: The empirical design is built from a large administrative panel and a within-product interest-rate comparison.

TABLE 1
Descriptive statistics across samples

	(1) Full sample mean/sd	(2) Refinancer mean/sd	(3) Refinancer panel mean/sd
Interest rate (%)	4.12 (1.29)	4.26 (1.38)	4.06 (1.23)
Loan size (£)	142,710.6 (120,512.8)	138,376.2 (120,725.2)	145,400.8 (116,489.4)
Property value (£)	257,004.0 (255,284.2)	263,076.4 (270,122.1)	262,304.0 (247,074.1)
Loan-to-value ratio (%)	59.9 (22.1)	56.4 (21.7)	58.8 (19.8)
Gross income (£)	57,270.6 (84,069.3)	57,480.9 (84,517.4)	57,483.2 (77,513.0)
Loan-to-income ratios	2.78 (1.85)	2.68 (2.03)	2.78 (1.60)
Repayments to income (%)	21.8 (39.5)	22.6 (48.4)	21.9 (15.4)
Proportion with joint income	0.54 (0.50)	0.55 (0.50)	0.55 (0.50)
Mortgage term (years)	20.7 (7.44)	19.0 (6.92)	20.0 (6.65)
Time to refinance (years)	2.80 (1.71)	2.80 (1.70)	3.27 (1.72)
Proportion of fixed rate mortgages	0.68 (0.47)	0.65 (0.48)	0.73 (0.45)
Proportion of refinance events	0.56 (0.50)	0.88 (0.32)	0.76 (0.42)
Borrower's age	40.0 (10.0)	41.7 (9.57)	40.3 (8.93)
Observations	3,049,164	1,961,325	647,192

(a) Descriptive statistics across samples.

entering into individual negotiations that depend on idiosyncratic factors.

Our empirical analysis will be based on the average interest rate jump at each notch conditional on bank, product, and time. We estimate these interest rate jumps non-parametrically using the following regression:

$$r_i = f(\text{LTV}_i) + \beta_1 \text{bank}_i + \beta_2 \text{variability}_i \otimes \text{duration}_i \otimes \text{month}_i + \beta_3 \text{repayment}_i + \beta_4 \text{term}_i + v_i, \quad (2.1)$$

where r_i is the nominal mortgage interest rate for individual i , $f(\cdot)$ is a step function with steps at

(b) Regression for conditional interest-rate jumps.

9.3 Figures 3–4: interest-rate jumps and counterfactual distribution

Read:

- Figure 3 shows that rates are nearly flat between notches and jump sharply at thresholds.
- The largest jumps are around 80% and 85% LTV.
- Figure 4A shows that passive LTVs are much smoother than actual LTVs.
- Figure 4B shows the adjusted counterfactual distribution used to measure bunching and missing mass.

Economic message: The paper separately identifies the incentive schedule and the distribution that would prevail without the notches.

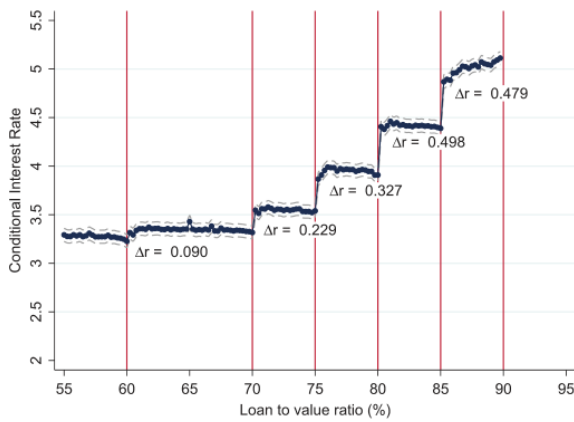


FIGURE 3

Interest rate jumps at notches

shows the conditional interest rate as a function of the LTV ratio from the non-parametric regression coefficient on the LTV bin dummy plus a constant given by the mean predicted value $E(\hat{r}_i)$ from a

(a) Estimated interest-rate jumps at notches.

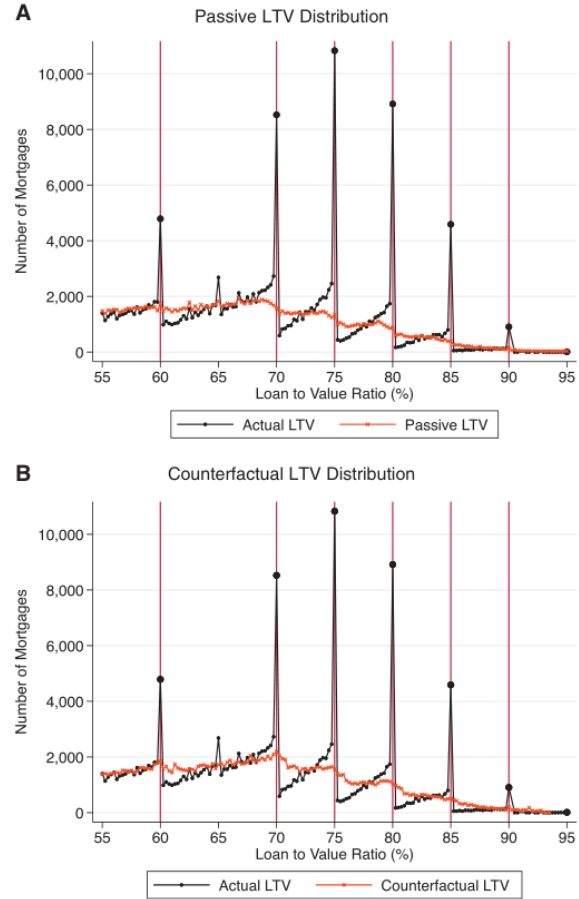


FIGURE 4

Constructing the counterfactual LTV distribution.

(b) Actual, passive, and counterfactual LTV distributions.

9.4 Figure 5 and simple-model equations

Read:

- Figure 5 gives the core bunching logic in a two-period choice diagram and a density diagram.
- The marginal buncher is indifferent between locating at the notch and locating above it.
- Equations (3.5)–(3.7) eliminate unobserved wealth and compute utility at the notch and the best interior point.

Economic message: The structural mapping from bunching to EIS is an indifference condition for the marginal buncher.

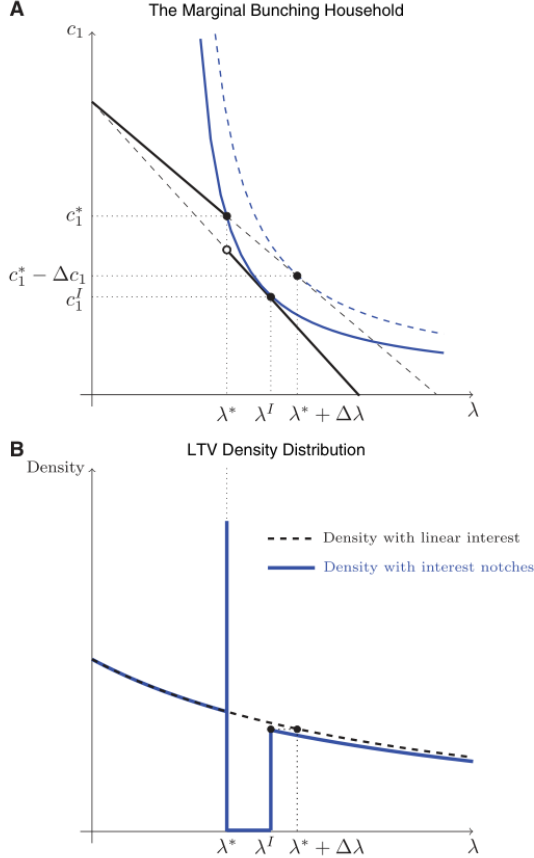


FIGURE 5
Bunching in a simple model of intertemporal substitution.
(a) Bunching in the simple model.

$$W_0 = P_0 H - y_0 + \frac{y_1 + (1-d)P_1 H - ((\delta R)^\sigma + R)(\lambda^* + \Delta\lambda)P_0 H}{(\delta R)^\sigma}. \quad (3.5)$$

This relationship allows us to eliminate W_0 from the problem. This is helpful because our data do not contain information on non-housing assets and liabilities, and therefore do not enable us to measure total initial wealth.

Using wealth defined in equation (3.5) and the optimality conditions (3.2)–(3.4) evaluated at the interest rate $R + \Delta R$, we can solve for the lifetime utility of the marginal buncher at the best interior choice λ^I in the presence of the notch. This is given by

$$V^I(\sigma, \delta, \Delta\lambda, \Delta R, \mathbf{x}) = \frac{\sigma}{\sigma-1} (P_0 H)^{\frac{\sigma-1}{\sigma}} \frac{(\delta^\sigma (R + \Delta R)^{\sigma-1} + 1)^{\frac{1}{\sigma}}}{(\delta R)^{\sigma-1}} \times \left(\left(\frac{(\delta R)^\sigma}{R + \Delta R} + 1 \right) \left(\frac{y_1}{P_0 H} + \Pi_1 \right) - ((\delta R)^\sigma + R)(\lambda^* + \Delta\lambda) \right)^{\frac{\sigma-1}{\sigma}}, \quad (3.6)$$

where $\Pi_1 \equiv (1-d)\frac{P_1}{P_0}$ is gross house price growth net of depreciation. In the indirect utility function $V^I(\cdot)$, the argument $\mathbf{x}$ is a vector that includes the parameters $\{\lambda^*, R, \frac{y_1}{P_0 H} + \Pi_1\}$.

Similarly, setting $\lambda = \lambda^*$ and applying the interest rate R , the budget constraints (3.2) and (3.3) and the wealth condition (3.5) allow us to evaluate lifetime utility at the notch as

$$V^N(\sigma, \delta, \Delta\lambda, \mathbf{x}) = \frac{\sigma}{\sigma-1} (P_0 H)^{\frac{\sigma-1}{\sigma}} \left(\frac{1}{(\delta R)^\sigma} \left(\frac{y_1}{P_0 H} + \Pi_1 - R\lambda^* - ((\delta R)^\sigma + R)\Delta\lambda \right)^{\frac{\sigma-1}{\sigma}} \right). \quad (3.7)$$

(b) Key wealth and indirect-utility equations.

9.5 Propositions 1–3: structural mapping, borrowing elasticity, and frictions

Read:

- Proposition 1 defines the estimating indifference equation for the EIS.
- Proposition 2 shows how the structural EIS maps into the reduced-form borrowing elasticity.
- Proposition 3 characterizes the dominated region above a notch.

Economic message: The paper does not simply report bunching. It explains exactly how bunching, borrowing elasticities, and optimization frictions relate to the structural EIS.

Proposition 1 (Estimating indifference equation) Given a bunching moment $\{\Delta\lambda, \Delta R\}$ and a discount factor δ , the EIS σ is the solution to the indifference equation

$$F(\sigma, \delta, \Delta\lambda, \Delta R, \mathbf{x}) \equiv V^N(\sigma, \delta, \Delta\lambda, \mathbf{x}) - V^I(\sigma, \delta, \Delta\lambda, \Delta R, \mathbf{x}) = 0, \quad (3.8)$$

where $\mathbf{x} = \left\{ R, \lambda^*, \frac{y_1}{p_0^1 \delta} + \Pi_1 \right\}$, and where $V^I(\cdot)$ and $V^N(\cdot)$ are given by (3.6) and (3.7), respectively.

Proof. The proof is in [Supplementary Appendix B](#). $\square$

Three points are worth highlighting. First, the indifference equation (3.8) is based on a setting with only one notch, while our empirical setting has multiple notches. In the presence of multiple notches, it is possible that bunchers move across more than one threshold at a time, and it is conceptually straightforward to modify the indifference equation to allow for this (see [Kleven and Waseem, 2013](#)). We focus on the single-notch equation here, because the data does not support the presence of multiple-notch jumps. We discuss this point in the next section.

Second, the indifference equation contains two structural parameters, the EIS σ and the discount factor δ . This suggests that we cannot identify σ from a single bunching moment. However, it turns out that the value of σ is extremely robust to assumptions about δ . The intuitive reason is that the discount factor primarily governs the *level* of borrowing at any interest rate (*i.e.* it shifts the LTV distribution on both sides of the notch) and has only a minor impact on the *response* of borrowing to interest rate changes. In contrast, the EIS governs the curvature of intertemporal preferences, which directly impacts bunching responses. This is immediately apparent when differentiating the Euler equation (3.4):

$$\frac{\partial \log(c_1/c_0)}{\partial \log R} = \sigma. \quad (3.9)$$

(a) Proposition 1: estimating indifference equation.

as we will show, it is a general feature of our methodology that the calibrated variables have a very small impact on the estimating indifference equation, making our results robust despite the analytical complexity of the expressions. The intuitive reason is essentially the same as the one underlying the robustness to δ .

The exposition above assumes that there is only one value of the structural EIS σ , while in practice there is likely to be heterogeneity in this parameter. In fact, the empirical LTV distribution shown in [Figure 1](#) implies that this has to be the case: without heterogeneity, there would be a *sharp* hole in the LTV distribution between λ^* and λ^l as illustrated in [Figure 5B](#), whereas the empirical LTV distribution features a gradual hole and has some refinancers located just above the notch.¹⁹ This provides *prima facie* evidence that some households have very small σ s while others have larger σ s. As [Kleven and Waseem \(2013\)](#) and [Kleven \(2016\)](#) show, in the presence of heterogeneity in σ , our bunching approach estimates the average σ .

To see this, consider a joint distribution of initial wealth W_0 and the EIS σ . At each elasticity level σ , households optimize as characterized above. In the counterfactual scenario with a constant interest rate R , there is a joint distribution of LTV and EIS given by $g_0(\lambda, \sigma)$ and an unconditional distribution of LTV given by $g_0(\lambda) = \int_{\sigma} g_0(\lambda, \sigma) d\sigma$. In the observed scenario with a notched interest rate, the empirical bunching at elasticity level σ produces LTVs for $\lambda \in \lambda^l, \lambda^*$. We use these data

(b) Proposition 2: EIS versus borrowing elasticity.

based on the presence of strictly dominated regions of behaviour above notches. In their setting, strictly dominated regions above income tax notches were used to estimate the fraction of non-optimizing agents, while being agnostic about the specific reasons for not optimizing. Assuming that the fraction of non-optimizers is the same outside the dominated region (*i.e.* where it cannot be directly measured), [Kleven and Waseem \(2013\)](#) showed that it is possible to adjust the bunching estimates for the amount of optimization friction in order to estimate true structural elasticities.

Here, we propose a parametric version of the Kleven–Waseem friction approach. In our setting, there are no strictly dominated regions per se. Locating immediately above an LTV notch implies a large drop in future consumption, but allows for (slightly) larger current consumption. If a consumer is perfectly impatient ($\delta = 0$), locating in such regions will be optimal. However, as long as consumers value future consumption at all ($\delta > 0$), there exists no non-negative EIS ($\sigma \geq 0$) that can justify locating immediately above an LTV threshold. Even with a zero substitution elasticity, the higher interest rate above the notch creates a wealth effect that should make consumers reduce

Proposition 3: dominated region above the notch.

9.6 Figures 6–8: simulated and empirical bunching

Read:

- Figure 6 shows that low EIS values fit the observed distribution better than high EIS values.
- Figure 7 pools all notches and reports raw bunching, adjusted bunching, and the pooled interest-rate jump.
- Figure 8 repeats the estimation separately for the 70%, 75%, 80%, and 85% notches.
- Bunching rises with the interest-rate jump.

Economic message: These figures are the empirical heart of the paper: they show both the behavioral response and why that response implies low intertemporal substitutability.

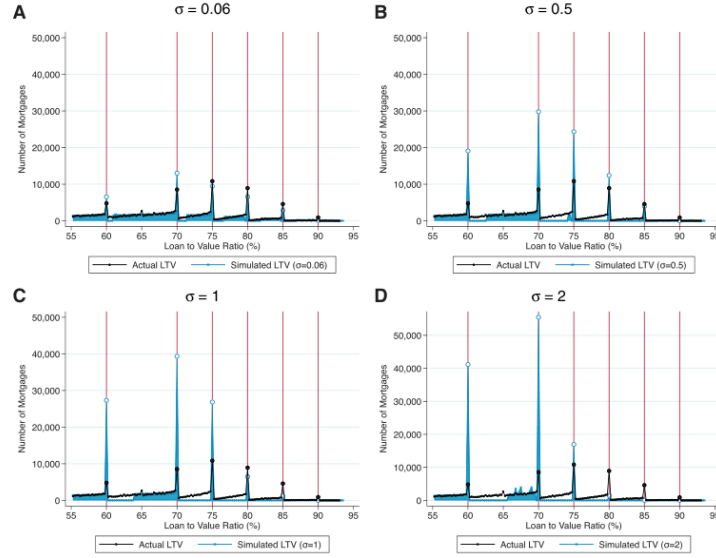


FIGURE 6

Observed versus simulated LTV distributions when varying the EIS.

Notes: The figure shows simulations of a model introduced in Section 3 for a range of EIS values. The lighter lines show the predicted

Observed versus simulated LTV distributions for different EIS values.

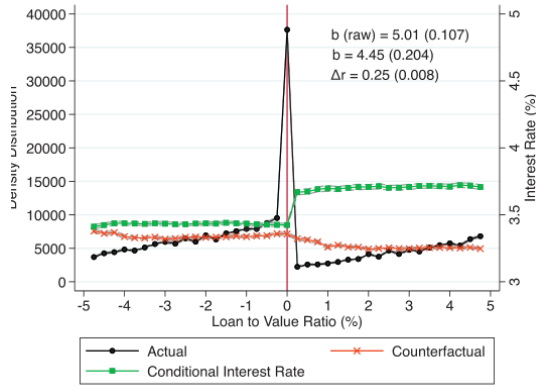


FIGURE 7

Bunching estimation when pooling notches

ws the actual ($f(\lambda)$) with black dots and counterfactual ($f_0(\lambda)$) with crosses) distributions of 5%, 80%, and 85%). The squares show the conditional interest rate in each LTV bin from the footnote to Figure 3. The counterfactual LTV distribution is obtained using the method outline

(a) Pooled notches.

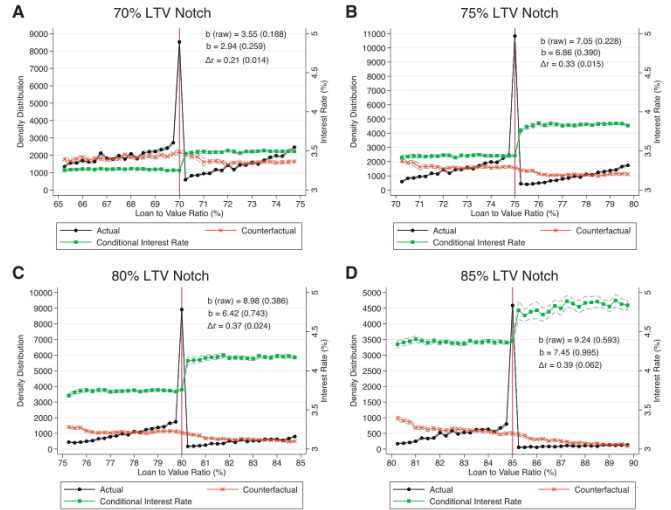


FIGURE 8

Bunching estimation at individual notches

(b) Individual notches.

9.7 Tables 2–4: simple-model estimates, friction bounds, and heterogeneity

Read:

- Table 2 estimates the EIS from individual and pooled notches in the simple model.
- The pooled EIS is 0.07; individual estimates range from 0.03 to 0.17.
- Table 3 shows that even extreme friction assumptions only raise the pooled EIS to about 0.30.

- Table 4 shows modest heterogeneity by age, income, leverage, income growth, house price growth, and passive interest-rate changes.

Economic message: The low EIS is not an artefact of one threshold, one friction adjustment, or one borrower subgroup.

TABLE 2
From bunching to the EIS: simple model

Statistic	Notch				
	60	70	75	80	85
Panel A: Bunching evidence					
$r(\%)$	3.17 (0.01)	3.25 (0.00)	3.44 (0.00)	3.76 (0.00)	4.38 (0.01)
$\Delta r(\%)$	0.10 (0.01)	0.21 (0.01)	0.33 (0.02)	0.37 (0.02)	0.39 (0.06)
b	0.96 (0.17)	2.94 (0.26)	6.86 (0.39)	6.42 (0.74)	7.45 (0.99)
a	0.58 (0.05)	0.21 (0.02)	0.30 (0.03)	0.15 (0.02)	0.08 (0.03)
b_{Adj}	2.31 (0.49)	3.73 (0.35)	9.87 (0.60)	7.59 (0.89)	8.11 (1.16)
$\Delta \lambda_{Adj}$	0.67 (0.14)	1.06 (0.09)	3.32 (0.18)	2.68 (0.32)	3.71 (0.70)
Panel B: Elasticities					
EIS σ	0.03 (0.01)	0.03 (0.00)	0.17 (0.02)	0.08 (0.02)	0.13 (0.05)
Reduced-form ε	0.53 (0.01)	0.53 (0.00)	0.60 (0.01)	0.56 (0.01)	0.58 (0.02)

Notes: The table shows our reduced-form estimates using bunching at the various LTV notches separately, and pooling the notches from 60% to 85%. r is the conditional nominal interest rate below the notch, Δr is the interest rate jump at the notch, estimated as described in Section 2.3. b_{Adj} is our normalized bunching estimate, net of round number bunching.

(a) From bunching to the EIS: simple model.

TABLE 3
Bounding optimization frictions and the EIS

	Notch			
	70	75	80	85
Panel A: Adjustment factor a				
(1) Notched banks only	0.11 (0.02)	0.15 (0.02)	0.15 (0.03)	0.03 (0.01)
(2) Dominated region	0.21 (0.02)	0.30 (0.03)	0.15 (0.02)	0.08 (0.03)
(3) Entire hole	0.67 (0.05)	0.60 (0.02)	0.57 (0.04)	0.40 (0.09)
Panel B: Elasticity of intertemporal substitution σ				
(4) Unadjusted	0.02 (0.00)	0.08 (0.01)	0.06 (0.01)	0.11 (0.04)
(5) Dominated region: notched banks only	0.02 (0.00)	0.11 (0.01)	0.08 (0.02)	0.11 (0.04)
(6) Dominated region: all banks	0.03 (0.00)	0.17 (0.02)	0.08 (0.02)	0.13 (0.05)
(7) All mass in the hole is friction	0.16 (0.05)	0.50 (0.07)	0.31 (0.08)	0.30 (8.53)

Notes: The table shows how the estimated EIS is affected by assumptions on optimization frictions. The top panel of the table shows the friction adjustment factor a estimated in three different cases. Row (1) shows the friction adjustment based on mass in the dominated region using only notched banks, row (2) shows the friction adjustment based on mass in the dominated region using all banks (our baseline estimates), while row (3) shows the friction adjustment assuming that all mass in the hole is due to friction. The bottom panel of the table shows the estimated EIS when not adjusting for optimization friction (in row (4)), and when adjusting for friction using each of the three measures provided in the top panel (in rows (5)–(7)). As explained in the main text of the article, the EIS estimates provided in rows (4) or (5) are in general lower bounds, whereas the EIS estimate provided in row (7) is an upper bound. The upper bound is based on the extreme assumption that all density mass in the hole—not just the mass in the much narrower dominated region—can be explained by friction rather than by heterogeneity in true preferences ($\bar{U} \neq \bar{U}^*$). True preferences are assumed to be homogeneous.

(b) Bounds from optimization-friction assumptions.

TABLE 4
Heterogeneity in the EIS

Covariate	Quartile			
	1	2	3	4
Age	0.05 (0.01)	0.09 (0.02)	0.10 (0.02)	0.15 (0.08)
Household income	0.09 (0.02)	0.08 (0.01)	0.07 (0.01)	0.05 (0.01)
Loan-to-income	0.02 (0.01)	0.05 (0.01)	0.08 (0.01)	0.07 (0.02)
Income growth	0.05 (0.01)	0.06 (0.02)	0.07 (0.01)	0.07 (0.02)
House price growth rate	0.06 (0.02)	0.05 (0.01)	0.04 (0.01)	0.13 (0.03)
Interest rate change (passive)	0.02 (0.01)	0.06 (0.02)	0.11 (0.03)	0.11 (0.03)

Notes: The table shows the heterogeneity in our estimated EIS σ (using the pooled average notch) by age, income, loan to income (LTI), income growth, house price growth, interest rate change since the previous mortgage (assuming passive borrower behaviour). For each covariate, we partition the refinancer panel into four quartiles and separately estimate σ in each quartile. The standard errors, shown in parentheses, are obtained by bootstrapping the estimation routine, stratifying by notch, 100 times.

Heterogeneity in the EIS.

9.8 Full structural model and robustness

Read:

- The full model adds housing services, liquid assets, uncertainty, risk aversion, bequest motives, refinance fees, amortization, and mortgage reset rates.
- Table 5 shows that the full structural model gives an average EIS of 0.08.
- Table 6 varies discounting, present bias, risk aversion, future rates, house price trends, house price variance, income profiles, unemployment probabilities, and replacement rates.
- The estimates remain low across specifications.

Economic message: The lifecycle model confirms that the simple-model estimate is not driven by unrealistic assumptions about risk, liquidity, or dynamics.

$$V_t = \left(\left(c_t^\sigma H_{t+1}^{1-\alpha} \right)^{\frac{\sigma-1}{\sigma}} + \delta \left(E_t \left[V_{t+1}^{1-\gamma} \right] \right)^{\frac{1}{1-\gamma} \frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \quad (5.13)$$

where V_t is the net present value of utility at period t , σ is the EIS, γ is the coefficient of relative risk aversion, and α is the share of housing in overall consumption. These preferences follow Epstein and Zin (1989) and Weil (1990) and allow for a distinction between risk aversion and the EIS. This helps ensure that our EIS estimates are not influenced by the dual role this parameter plays in standard CRRA preferences. The unit elasticity of substitution between housing and non-housing consumption (Cobb–Douglas) is justified by the observation that expenditures on housing services have historically been a constant share of total household expenditures (see e.g. Piazzesi and Schneider, 2016).

Households value end-of life wealth W_{T+1} via a reduced-form bequest motive, i.e.

$$U_{T+1} = \gamma W_{T+1} \quad (5.14)$$

(a) Full model: recursive preferences and bequest value.

With these features, the household’s budget constraint is given by

$$\begin{aligned} c_t &= y_t + (1 - \pi_t)L_t - L_{t+1} \\ &\quad + P_t((1-d)H_t - H_{t+1}) \\ &\quad + D_{t+1} - R_t D_t - \Omega_t^R, \end{aligned} \quad (5.15)$$

where I_t^R is an indicator equal to 1 if the household refinances ($D_{t+1} \neq (1 - \mu_t)D_t$), and π_t gives

(b) Full model: budget constraint.

TABLE 5
From bunching to the EIS: full structural model

Statistic	Notch				
	60	70	75	80	Average
b	0.96 (0.17)	2.94 (0.26)	6.86 (0.39)	6.42 (0.74)	7.45 (0.99)
a	0.58 (0.05)	0.21 (0.02)	0.30 (0.03)	0.15 (0.02)	0.08 (0.03)
b_{Adj}	2.31 (0.49)	3.73 (0.35)	9.87 (0.60)	7.59 (0.89)	8.11 (1.16)
$\Delta\lambda_{Adj}$	0.67 (0.14)	1.06 (0.09)	3.32 (0.18)	2.68 (0.32)	3.71 (0.70)
EIS σ	0.05 (0.01)	0.04 (0.01)	0.11 (0.01)	0.11 (0.02)	0.28 (0.15)

Notes: The table shows the results of our structural estimation as described in Section 5. b is our normalized bunching estimate as described in Section 3 and the footnote to Figure 8. a is the adjustment factor for optimization frictions (the number of individuals observed in the dominated region divided by the number of individuals in the same region in the counterfactual distribution), and $b_{Adj} = b/(1-a)$ is our bunching estimate, adjusted for optimization frictions. $\Delta\lambda$ is the leverage response corresponding to b_{Adj} . σ is the EIS that solves the full model described in Section 5. The standard errors, shown in parentheses, are obtained by bootstrapping the estimation routine, stratifying by notch, 100 times.

(a) Full structural model estimates.

TABLE 6
Robustness of EIS estimates

(1)	Discount factor δ	0.7	0.9	0.96	0.99
		0.13 (0.015)	0.12 (0.015)	0.08 (0.011)	0.12 (0.013)
(2)	Present bias factor β	0.3	0.5	0.7	1
		0.17 (0.026)	0.14 (0.019)	0.11 (0.015)	0.08 (0.011)
(3)	Risk aversion γ	0	1	2	CRRA
		0.11 (0.011)	0.12 (0.012)	0.08 (0.011)	0.15 (0.013)
(4)	Future interest rates	+0pp	+1pp	+2pp	+3pp
		0.08 (0.011)	0.11 (0.014)	0.12 (0.013)	0.12 (0.013)
(5)	House price trend	−0.6%	0	0.6%	6%
		0.10 (0.014)	0.10 (0.014)	0.08 (0.011)	0.10 (0.014)
(6)	House price variance	0	0.004	0.006	0.008
		0.08 (0.009)	0.10 (0.011)	0.08 (0.011)	0.16 (0.012)
(7)	Lifecycle income profile	Peak Slope	£44K 0%	£46K 0.7%	£56K 2.7%
			0.12 (0.016)	0.08 (0.011)	0.09 (0.008)
(8)	Unemployment probability	3%	5%	7%	10%
		0.09 (0.014)	0.08 (0.011)	0.11 (0.013)	0.12 (0.015)
(9)	Replacement rate	60%	80%	100%	
		0.08 (0.011)	0.13 (0.013)	0.12 (0.016)	

Notes: The table shows the robustness of our estimates of the EIS σ to a number of the assumptions of our structural model. Panel (1) varies the discount factor δ . Panel (2) relaxes our assumption that households discount the future with traditional geometric discounting and allows for quasi-hyperbolic $\beta - \delta$ discounting with present bias parameters β from 0.3 to 1 (geometric discounting). Panel (3) varies the coefficient of relative risk aversion γ . Panel (4) varies (deterministic) future interest rates by shifting the entire yield curve up by 1pp to 3pp. Panel (5) varies real house price growth ranging

(b) Robustness of EIS estimates.

10 Short summary

- The paper estimates the EIS using U.K. mortgage interest-rate notches.
- U.K. mortgage rates jump discretely at LTV thresholds, creating incentives to reduce borrowing just below those thresholds.
- Administrative mortgage data show sharp bunching below notches and missing mass above notches.
- The authors focus on refinancers to isolate borrowing from housing demand.
- They construct a counterfactual LTV distribution using passive LTVs from the prior mortgage and equity-extraction adjustments.
- A simple two-period model maps bunching into the EIS through a marginal-buncher indifference condition.
- The model also derives why reduced-form borrowing elasticities are not the same as the EIS.
- The paper corrects for optimization frictions using dominated regions above notches.

- The simple model estimates an average EIS around 0.07–0.1.
- The full stochastic lifecycle model estimates an average EIS around 0.08.
- Robustness checks do not generate EIS values near one.
- The incremental contribution is a quasi-experimental, bunching-based structural estimate of a central macro-finance preference parameter.

11 Conclusion

11.1 Core conclusion

Best, Cloyne, Ilzetzki, and Kleven show that mortgage-rate notches can be used to estimate the elasticity of intertemporal substitution. Their central empirical finding is that the household EIS is small, around 0.1. This result is robust across a simple structural model, a rich stochastic lifecycle model, friction adjustments, and heterogeneity analyses.

11.2 Economic intuition

Borrowers visibly respond to mortgage-rate notches by bunching below LTV thresholds. But the incentives are so large that a high EIS would imply much more dramatic movements: households would jump across multiple notches and leave large holes in the LTV distribution. Because that does not happen, the implied EIS is low. The observed borrowing response is driven largely by the wealth effect of higher mortgage payments rather than by strong intertemporal substitution.

11.3 Incremental takeaway

The paper’s main incremental contribution is to connect quasi-experimental bunching evidence to a deep preference parameter in a dynamic setting. It moves beyond both traditional Euler-equation EIS estimation and reduced-form mortgage-borrowing elasticities. It also extends bunching methods by using dominated regions above notches to discipline optimization frictions. The resulting evidence challenges calibrations that rely on large EIS values and suggests that interest-rate policy may operate more through cash-flow and wealth channels than through intertemporal substitution.